

INVITED TALK / ABSTRACT

A non local PDE model for fire fronts

Valentina Wheeler

University of Wollongong

Abstract

In this talk I will introduce a new model for describing front propagation in bush fires. The model describes the temperature diffusion in view of an ignition process induced by an interaction kernel and the effects of environmental wind and fire wind. This is a hybrid model incorporating both the geometry of the front and line interactions. I will present the idea behind the model, some of the analysis and special solutions with their applications. This is joint work with Serena Dipierro, Enrico Valdinoci and Glen Wheeler.

Keywords: non local PDE model; fire fronts; front propagation; bush fires; temperature diffusion; ignition process; interaction kernel; environmental wind